INCH-POUND

MIL-C-60111C
AMENDMENT 5
28 February 1995
SUPERSEDING
AMENDMENT 4
28 February 1994

MILITARY SPECIFICATION

CARTRIDGE, 5.56MM, TRACER, M196

This amendment forms a part of MIL-C-60111C (AR), dated 15 October 1976, and is approved for use by the US Army Armaments, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, Delete in its entirety and substitute the following:

"1.1 This specification covers Cartridge, 5.56mm, Tracer, M196 for use in the M16 and M16A1 Rifles and the M231 Port Firing Weapon."

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2.1, Delete:

"10534605 - Packing and Marking, Cartridges, 5.56mm, Cartons; Bandoleers; Box, Ammunition, M2A1; Box, Wirebound"

Substitute:

"12551963 - Packing and Marking for Box Wirebound for Cartridge 5.56mm"

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2.1, Delete the following:

"10535858 - Packing and Marking, Cartridges, 5.56mm; Cartons; Box, Ammunition, M2A1; Box, Wirebound

10542258 - Packing and Marking, Cartridges, 5.56mm; 10 Rd. Clips; Cartons; Bandoleers; Box, Ammunition, M2A1; Box, Wirebound

11735709 - Packing and Marking, Cartridges, 5.56mm; 10 Rd. Clips; Cartons; Bandoleer M8; Box, Ammunition, M2A1; Box, Wirebound"

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3.12, Delete in its entirety and substitute the following:

"3.12 Function and casualty. The cartridge shall function without casualty at ambient temperature and under the conditions specified in 3.11 when fired in the M16 or M16A1 Rifles and the M231 Port Firing Weapon."

3.14, Delete in its entirety and substitute the following:

"3.14 Fouling. The fouling accumulated in the weapon during the firing of 1000 sample cartridges shall not cause failure of the weapon to function when fired in the M16 or M16A1 Rifles and the M231 Port Firing Weapon."

Add the following paragraph:

"3.15.3 Bullet tip identification. Each cartridge shall be marked on the bullet tip with the appropriate color of paint in accordance with the applicable drawing(s). Prior to linking, clipping or insertion into cartons, cartridges missing up to 10% of the required bullet tip identification shall be acceptable. Following any necessary handling or orientation and after linking or clipping or prior to insertion into cartons, cartridges missing up to 40% of the required bullet tip identification shall be acceptable."

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4.2.1, Delete in its entirety and substitute the following:

"4.2.1 Initial production sample. At the beginning of regular production, a sample of 5200 cartridges shall be submitted in accordance with contract requirements. In

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addition, 100 bullets shall be submitted for visual and dimensional examinations. The sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply."

* 4.3.1 Add the following new subparagraph:

"4.3.1.3 Components parts. Unless otherwise specified, component lots shall be homogeneous and of a size convenient to the contractor and inspected, tested and accepted by the contractor. The cartridge lot shall not contain:

a. Cartridge cases from more than one manufacturer.
b. Primers from more than one lot interfix number from one manufacturer.
c. Bullets from more than one manufacturer.
d. Bullets from more than one process.
e. Propellant from more than two lots.
f. Propellent from more than one manufacturer.

4.3.2, Delete in its entirety and substitute the following:

"4.3.2 Examination. 100% examination shall be performed for all critical and special defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 0.25% and the AQL for the minor class shall be 1.5%. Examination shall be visual or gage and weigh or by means of Government approved automated inspection system such as optical, mechanical or electrical. All non-conforming cartridges shall be rejected. If a visual critical defect is found in a sample either just prior to a firing test or after a firing test (and the defect is not due to the firing) the lot shall be rejected. The lot shall be rescreened and resubmitted for visual inspection of critical defects. If a critical defect is found during packing, the portion of the lot that has been packed or is in the process of being packed shall be rejected. In addition the portion of the lot remaining to be packed shall be rejected. The lot shall then be rescreened and resubmitted for visual inspection of critical defects."

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Table 1, add the following:

"3. Incomplete or missing tip identification (red) 3/, Minor x"

* Delete characteristic 6 in its entirety and substitute the following:

"No.	Defect and Method of Inspection	Critical	Special	Major	Minor	Major or Minor
6	Split case In I,S, or J location when it is determined that there is no potential for loss of propellant		X			
	In I,S, or J location when it is determined that there is a potential for loss of propellant			X		
	In K, L, or M location	X"				

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Delete Characteristics 32, 33 and 34 in their entirety and substitute the following:

"No.	Defect and Method of Inspection	Critical	Special	Major	Minor	Major or Minor
32.	No Primer		X4/			
33.	Cocked Primer		X4/			
34.	Inverted Primer		X4/"			

Delete characteristic 37 in its entirety and substitute the following:

"No.	Defect and Method of Inspection	Critical	Special	Major	Minor	Major or Minor
37	Insufficient/smeared waterproofing material (primer pocket joint)"		X			

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Bottom of Table 1, add the following:

"3/ Reference paragraph 3.15.3"

"4/ Refer to MIL-A-48078, para 6.5.4, for the definition of a special defect:

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4.3.3.1 Table II, Delete in its entirety and substitute the following:

TABLE II

Test	Number of Cartridges				
	Ambient Temp	70 ± 2°F	Low Temp (see 4.4.10.1)	High Temp (see 4.4.10.2)	Requirement Para
Bullet extraction 1/	25				3.2
Residual stress (mercurous nitrate) 1/	50				3.3
Airtightness of base base closure seal 2/	--				3.5
Waterproof 3/	50				3.4
Accuracy 4/	90				3.6
Trace 5/	100				3.7
Velocity 6/		20	20	20	3.8 & 3.11
Chamber pressure 6/		20	20	20	3.9 & 3.11
Port pressure 6/		--	--	--	3.10 & 3.11
Function and casualty 7/					
Rifle, M16 or M16A1			360	360	3.12 & 3.11
M231 Port Firing Weapon	360		360	360	3.12 & 3.11
Fouling 8/					
Rifle, M16 or M16A1	1000				3.14
M231 Port Firing Weapon	1000				3.14
Bullet Integrity 10/	100				3.13
Hardness					
Head 9/	10				Drawing
Sidewall 4/	10				Drawing

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4.3.4, Delete in its entirety and substitute the following:

"4.3.4 Packaging, packing and marking inspection. During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the lot does not contain a blank cartridge or a cartridge with the bullet missing. Examination shall be visual or by means of a Government-approved inspection system. All non-conforming cartridges shall be removed from the lot. A sample of 244 shall be selected from each lot and examined for the presence of a high pressure test, dummy or blank cartridge, or a cartridge with the bullet missing. If a high pressure test, dummy or blank cartridge, or a cartridge with the bullet missing is found, the lot shall be rejected. Any occurrence of a high pressure test, dummy or blank cartridge, or a cartridge with the bullet missing after this inspection shall cause the lot to be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing."

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Add new paragraph 4.3.6 as follows:

"4.3.6 Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.2). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable."

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4.4.11, Delete in its entirety and substitute the following:

"4.4.11 Function and casualty. This test shall be conducted in accordance with SCATP-5.56mm and 4.4.10 except that 30-round magazines loaded with 30 rounds each shall be used instead of the stated 20-round magazines. When the M231 Port Firing Weapon is used, 30-round bursts and short bursts (3-5 rounds) shall be fired from alternate magazines. Two M16 or M16A1 Rifles and Two M231 Port Firing Weapons shall be used and the following number of cartridges shall be fired in each:

	First Sample	Second Sample (if required)
Ambient Temperature	180	360
Low Temperature	180	360
High Temperature	180	360"

4.4.12, Delete in its entirety and substitute the following:

"4.4.12 Fouling. The test shall be conducted in accordance with SCATP-5.56 except that 30-round magazines loaded with 30 rounds each shall be used instead of the stated 20-round magazines. When the M231 Port Firing Weapon is used, the semi-automatic mode consists of short (3-5 rounds) bursts."

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5.1, Delete "F10535858, F105342258, F10544605, or F11735709." and replace with "12551963".

6.2: Change paragraph number from "6.2" to "6.3". Add the following paragraph:

"6.2 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity."

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Preparing activity:
Army-AR

(Project 1305-0109)

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